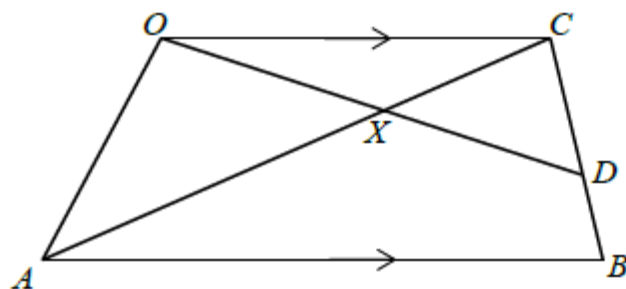


$$\overrightarrow{OP} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}, \overrightarrow{OQ} = \begin{pmatrix} -3 \\ -4 \end{pmatrix}, \overrightarrow{OR} = \begin{pmatrix} x \\ -1 \end{pmatrix}.$$

(a) Given that P , Q and R are collinear, find the value of x .

[3]

(b) The diagram show a trapezium $OABC$ such that $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$, $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$, $\overrightarrow{OC} = \frac{2}{3}\overrightarrow{AB}$, $\overrightarrow{BD} = \frac{3}{4}\overrightarrow{DC}$, and OD meets AC at X .



Find

(i) $\overrightarrow{OB}$,

[1]

(ii) $\overrightarrow{BD}$,

[2]

(iii) $\overrightarrow{OD}$.

[1]

(c) Given that $AX : AC = 9 : 13$, show that $\frac{OX}{OD} = \frac{7}{13}$.

[2]

(d) Find the ratio of the area of

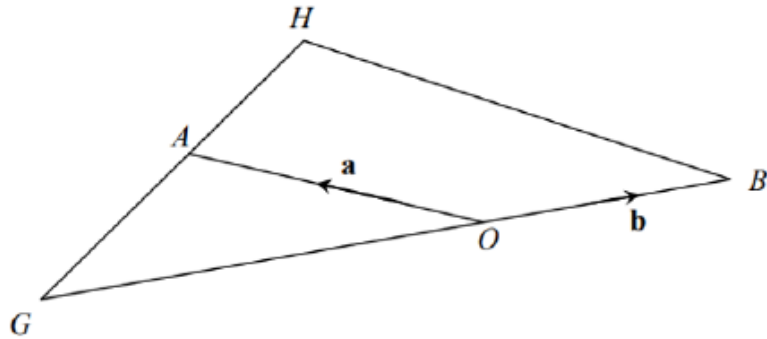
(i) triangle OAX to OXC ,

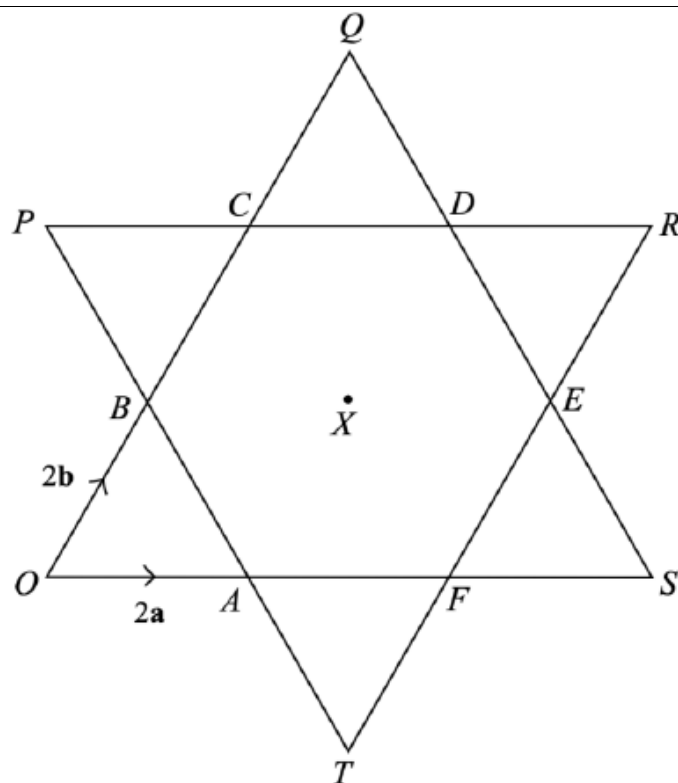
[1]

(ii) OAX to CXD .

[2]

2	SCGS/2025/I/26
	$OPQR$ is a rhombus. OST and PSQ are straight lines. $RQ = 2QT$. It is given that $\vec{OP} = 4\mathbf{p}$ and $\vec{OR} = 3\mathbf{r}$. (a) Express $\vec{ST}$, as simply as possible, in terms of $\mathbf{p}$ and $\mathbf{r}$. [1] (b) Using vectors, determine if OQ is parallel to PT. [4]
3	SCGS/2025/II/06
	In the diagram, P is the point $(0, 3)$ and T is the point $(8.5, 0)$. Q is a point such that $\vec{PQ} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$. (a) Find $\vec{PQ}$. [1] (b) Use vectors to find the position vector of Q. [2] $PQRS$ is a quadrilateral. The line SR intersects the x-axis at the point T and $\vec{SR} = \vec{PQ}$. (c) Find the equation of the line SR. [2] (d) Given that $ST : TR = 3 : 1$, find the coordinates of S. [2] (e) PS is perpendicular to SR. (i) Explain why $PQRS$ is a square. [2] (ii) Find the area of $PQRS$. [1]

4	CCHY/2025/I/25
	The position vectors of A and B, relative to O, are $\mathbf{a}$ and $\mathbf{b}$ respectively. $GB = 3OB$ and $GA = \frac{1}{2}GH$.  (a) Express each of the following in terms of $\mathbf{a}$ and $\mathbf{b}$. (i) $\overrightarrow{GA}$ [1] (ii) $\overrightarrow{HB}$ [2] (b) Use your answer in (a)(ii) to explain why $OAHB$ is not a trapezium. [2] (c) Find the ratio of (i) $\frac{\text{area of triangle } OHB}{\text{area of triangle } GHO}$, [1] (ii) $\frac{\text{area of triangle } OHA}{\text{area of } OAHB}$. [2]
5	CCHY/2025/II/06
	A is the point $(4, -2)$ and the line AB is parallel to the line $\frac{1}{2}x - 3y = 9$. (a) Find the equation of the line AB. [2] (b) The equation of the line BC is $2x + 9y + 52 = 0$. Find the x-coordinate of B. [2] (c) It is given that $\overrightarrow{AD} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$ and $\overrightarrow{AE} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}$. (i) By considering their magnitudes, show that ADE is a right-angled triangle. <i>Answer</i> [3] (ii) F is the point $(0, -6)$. Showing your working, determine if F lies on DA extended. [2]



A star is made up of a regular hexagon $ABCDEF$, centre X , surrounded by equilateral triangles.

$\overrightarrow{OA} = 2\mathbf{a}$ and $\overrightarrow{OB} = 2\mathbf{b}$.

- (a) Express the following vectors in terms of $\mathbf{a}$ and / or $\mathbf{b}$, giving your answers in the simplest form.

(i) $\overrightarrow{OR}$

[1]

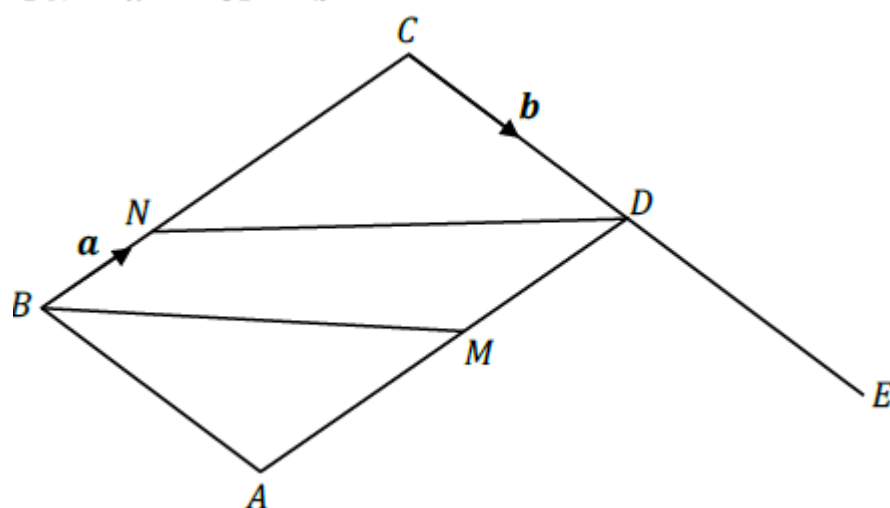
(ii) $\overrightarrow{FC}$

[1]

- (b) When $|\mathbf{a}| = 6$ units, find the perimeter of the star $OATFSERDQCPBO$.

[2]

$ABCD$ is a parallelogram, and E lies on CD produced such that $CD:CE = 1:2$.
 M is the midpoint of AD .
 N is a point on BC such that $BN:BC = 1:4$.
 $\overrightarrow{BN} = \mathbf{a}$ and $\overrightarrow{CD} = \mathbf{b}$.



(a) Express, as simply as possible, in terms of $\mathbf{a}$ and/or $\mathbf{b}$,

(i) $\overrightarrow{AM}$,

[1]

(ii) $\overrightarrow{BM}$,

[1]

(iii) $\overrightarrow{BE}$.

[1]

(b) Explain why B , M and E lie on a straight line.

[2]

(c) Find the numerical value of

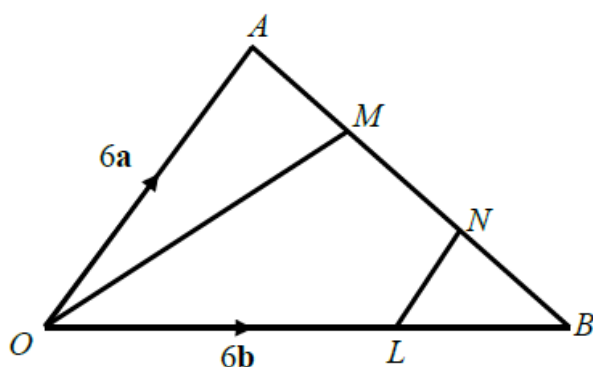
(i) $\frac{\text{Area of triangle } AMB}{\text{Area of triangle } DCN}$,

[1]

(ii) $\frac{\text{Area of triangle } EDM}{\text{Area of } DCBM}$.

[2]

(a)



OAB is a triangle.

L is the point on OB such that $OL : LB = 2 : 1$.

Points M and N lie on AB such that $AM = MN = NB$.

$\overrightarrow{OA} = 6\mathbf{a}$ and $\overrightarrow{OB} = 6\mathbf{b}$.

(i) Show that $\overrightarrow{OM} = 4\mathbf{a} + 2\mathbf{b}$.

[2]

(ii) Show that $\overrightarrow{LN} = k\overrightarrow{OA}$, where k is a constant to be found.

[2]

(b) A is the point $(1, 3)$ and B is the point $(-5, 2)$.

$\overrightarrow{XY} = \begin{pmatrix} -2 \\ m \end{pmatrix}$ and $\overrightarrow{XY}$ is parallel to $\overrightarrow{AB}$.

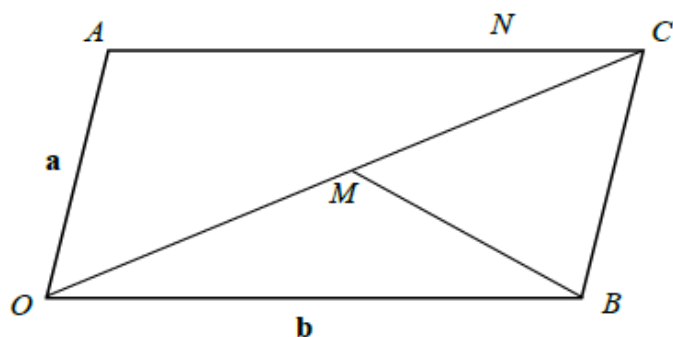
Find the value of m .

[3]

$OACB$ is a parallelogram.

M is the mid point of OC and $AN : AC = 3 : 4$.

$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$.



(a) Express, as simply as possible, in terms of $\mathbf{a}$ and $\mathbf{b}$,

(i) $\overrightarrow{BN}$,

[1]

(ii) $\overrightarrow{OC}$,

[1]

(iii) $\overrightarrow{BM}$.

[2]

(b) (i) Q is a point on OC such that $\overrightarrow{OQ} = \frac{3}{4}\mathbf{a} + \frac{13}{16}\mathbf{b}$.

Show that B , Q and N are collinear.

[2]

(ii) Find the ratio area triangle BCN : area triangle BCQ .

[1]

Solution

1

(a)

$$\overrightarrow{PR} = k\overrightarrow{RQ}$$

$$\begin{pmatrix} x \\ -1 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = k \left[\begin{pmatrix} -3 \\ -4 \end{pmatrix} - \begin{pmatrix} x \\ -1 \end{pmatrix} \right]$$

[M1] for collinear concept &
for position vectors

$$\begin{pmatrix} x-6 \\ -3 \end{pmatrix} = k \begin{pmatrix} -3-x \\ -3 \end{pmatrix}$$

$$k = 1$$

[M1] for finding k

$$x - 6 = -3 - x$$

$$2x = 3$$

$$x = \frac{3}{2}$$

[A1]

(bi)

$$\overrightarrow{AB} = \frac{3}{2}c$$

$$\overrightarrow{OB} = \frac{3}{2}c + a$$

(bii)

$$\overrightarrow{BC} = c - \left(\frac{3}{2}c + a \right)$$

$$= -\frac{1}{2}c - a$$

$$\overrightarrow{BD} = \frac{3}{7} \left(-\frac{1}{2}c - a \right)$$

$$= -\frac{3}{14}c - \frac{3}{7}a$$

(biii)

$$\overrightarrow{OD} = -\frac{3}{14}c - \frac{3}{7}a + \frac{3}{2}c + a$$

$$= \frac{4}{7}a + \frac{9}{7}c$$

(c)

		$\overrightarrow{OX} - \overrightarrow{OA} = \frac{9}{13}(c - a)$ $\overrightarrow{OX} = \frac{9}{13}c + \frac{4}{13}a$ $= \frac{1}{13}(4a + 9c)$ $= \frac{7}{13}\overrightarrow{OD}$ (di) $\frac{\text{Area of } OAX}{\text{Area of } OXC} = \frac{AX}{XC} = \frac{9}{4}$ (dii) $\frac{\text{Area of } OXC}{\text{Area of } CXD} = \frac{OX}{XD} = \frac{7}{6}$ $\frac{\text{Area of } OAX}{\text{Area of } OXC} \times \frac{\text{Area of } OXC}{\text{Area of } CXD}$ $= \frac{9}{4} \times \frac{7}{6}$ $= \frac{21}{8}$
2	(a) $\mathbf{r} + 2\mathbf{p}$ (b) $\overrightarrow{OQ} = 3\mathbf{r} + 4\mathbf{p}$ $\overrightarrow{PT} = \overrightarrow{PS} + \overrightarrow{ST} = 2\mathbf{r} + \mathbf{r} + 2\mathbf{p} = 3\mathbf{r} + 2\mathbf{p}$ Since $\overrightarrow{OQ}$ is not a scalar multiple of $\overrightarrow{PT}$ (or $\overrightarrow{OQ} \neq k \overrightarrow{PT}$ where k is a constant), OQ is not parallel to PT	
3	a $ \overrightarrow{PQ} = \sqrt{6^2 + 4^2}$ $= \sqrt{52}$ $= 7.21 \text{ units (3 s.f.)}$ b $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$ $\begin{pmatrix} 6 \\ 4 \end{pmatrix} = \overrightarrow{OQ} - \begin{pmatrix} 0 \\ 3 \end{pmatrix}$ $\therefore \overrightarrow{OQ} = \begin{pmatrix} 6 \\ 7 \end{pmatrix}$	

	c $m_{SR} = m_{PQ} = \frac{2}{3}$ Equation SR: $y - 0 = \frac{2}{3}(x - 8.5)$ $\therefore y = \frac{2}{3}x - \frac{17}{3}$
	d <u>Method 1</u> $\overrightarrow{ST} = \frac{3}{4}\overrightarrow{SR}$ $\overrightarrow{OT} - \overrightarrow{OS} = \frac{3}{4}\begin{pmatrix} 6 \\ 4 \end{pmatrix}$ $\begin{pmatrix} 8.5 \\ 0 \end{pmatrix} - \frac{3}{4}\begin{pmatrix} 6 \\ 4 \end{pmatrix} = \overrightarrow{OS}$ $\overrightarrow{OS} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} \therefore S(4, -3)$ d <u>Method 2</u> x-coord. of $S = 8.5 - \frac{6}{4}(3) = 4$ y-coord. of $S = 0 - \frac{4}{4}(3) = -3$ $\therefore S(4, -3)$
	ei Length of $PS = \sqrt{(0-4)^2 + (3+3)^2}$ $= 7.21$ units (3 s.f.) Since $\overrightarrow{PQ} = \overrightarrow{SR}$, PQ and SR are a pair of opposite sides that are parallel and equal. In addition, $\overrightarrow{PS} = \overrightarrow{PQ}$ and PS is perpendicular to SR. Thus, PQ is perpendicular to PS and $PS = SR$. i.e. QR will be perpendicular to PQ and SR and $QR = PQ = PS = SR$. $\therefore PQRS$ is a square. eii Area of $PQRS = \sqrt{52} \times \sqrt{52} = 52$ square units
4	(ai) $= \overrightarrow{GO} + \overrightarrow{OA} = 2\mathbf{b} + \mathbf{a}$ (aii) $\overrightarrow{GH} = 2(2\mathbf{b} + \mathbf{a})$ $\overrightarrow{HB} = \overrightarrow{HG} + \overrightarrow{GB}$ $= -4\mathbf{b} - 2\mathbf{a} + 3\mathbf{b}$ $= -\mathbf{b} - 2\mathbf{a}$ (b) Answer $\overrightarrow{OA}$ is not a scalar multiple of $\overrightarrow{HB}$, hence, OA is not parallel to HB and since AH is also not parallel to OB, thus, $OAHB$ is not a trapezium as it has no parallel sides. (ci)

$$= \frac{\frac{1}{2} \times OB \times h}{\frac{1}{2} \times GO \times h} = \frac{1}{2}$$

(cii)

$$\frac{\text{Area of } \triangle OHB}{\text{Area of } \triangle GHO} = \frac{1}{2}$$

$$\frac{\text{Area of } \triangle OHA}{\text{Area of } \triangle OHB} = \frac{1}{1}$$

$$\frac{\text{Area of } \triangle OHA}{\text{Area of } \triangle GHO} = \frac{1}{2}$$

$$\frac{\text{Area of } \triangle OHA}{\text{Area of } \triangle AHB} = \frac{1}{2}$$

5

(a)

$$\frac{1}{2}x - 3y = 9 \Rightarrow y = \frac{1}{6}x - 3$$

Since the lines are parallel, they share the same gradient.

$$-2 = \frac{1}{6}(4) + c \Rightarrow c = -\frac{8}{3}$$

The equation of line AB is $y = \frac{1}{6}x - \frac{8}{3}$.

(b)

$$2x + 9\left(\frac{1}{6}x - \frac{8}{3}\right) + 52 = 0$$

$$2x + \frac{3}{2}x - 24 + 52 = 0$$

$$\frac{7}{2}x = -28$$

$$x = -8$$

(ci)

$$AD^2 = \left| \overrightarrow{AD} \right|^2 = 3^2 + 3^2 = 18$$

$$AE^2 = \left| \overrightarrow{AE} \right|^2 = 5^2 + 1^2 = 26$$

$$\overrightarrow{DE} = \begin{pmatrix} 5 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

$$DE^2 = \left| \overrightarrow{DE} \right|^2 = 2^2 + (-2)^2 = 8$$

Since $AD^2 + DE^2 = 18 + 8 = 26$, we have $AD^2 + DE^2 = AE^2$.

By the converse of Pythagoras' Theorem, ADE is a right-angled triangle.

(cii)

	Since $\overrightarrow{AD} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$, the gradient of the line DA is $\frac{3}{3} = 1$. $-2 = 1(4) + c \Rightarrow c = -6$ The equation of the line DA is $y = x - 6$. Since the point F is the y-intercept of the line DA, it lies on DA extended.
6	(ai) $\overrightarrow{OR} = \overrightarrow{OA} + \overrightarrow{AF} + \overrightarrow{FE} + \overrightarrow{ER}$ $\overrightarrow{OR} = 4\mathbf{a} + 4\mathbf{b}$ (aii) $\overrightarrow{FC} = \overrightarrow{FE} + \overrightarrow{ER} + \overrightarrow{RD} + \overrightarrow{DC}$ $\overrightarrow{FC} = 4\mathbf{b} - 4\mathbf{a}$ (b) Perimeter of the star $= 12 (2 \times 6)$ $= 144$ units
7	(ai) $\overrightarrow{AM} = \frac{1}{2}\overrightarrow{AD} = \frac{1}{2}\overrightarrow{BC} = \frac{1}{2}(4\mathbf{a})$ (ABCD is a parallelogram) $\overrightarrow{AM} = 2\mathbf{a}$ (aii) $\overrightarrow{BM} = \overrightarrow{BA} + \overrightarrow{AM}$ $\overrightarrow{BM} = 2\mathbf{a} + \mathbf{b}$ (aiii) $\overrightarrow{BE} = \overrightarrow{BC} + \overrightarrow{CE}$ $\overrightarrow{BE} = 4\mathbf{a} + 2\mathbf{b}$ (b) $\overrightarrow{BM} = 2\mathbf{a} + \mathbf{b}$ $\overrightarrow{BE} = 4\mathbf{a} + 2\mathbf{b} = 2(2\mathbf{a} + \mathbf{b}) = 2\overrightarrow{BM}$ Since $\overrightarrow{BE} = 2\overrightarrow{BM}$ and B is a common point, B, M and E lie on a straight line. (Shown) (ci)

ABCD is a parallelogram, let h be the perpendicular height between AD and BC.

$$\frac{\text{Area of triangle } AMB}{\text{Area of triangle } DCN} = \frac{\frac{1}{2} \times h \times AM}{\frac{1}{2} \times h \times NC}$$

$$= \frac{AM}{NC} = \frac{2}{3}$$

(cii)

$$\frac{\text{Area of triangle } EDM}{\text{Area of triangle } ECB} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\frac{\text{Area of triangle } EDM}{\text{Area of } DCBM} = \frac{1}{3}$$

8

8ai

$$\begin{aligned}\overrightarrow{OM} &= \overrightarrow{OA} + \overrightarrow{AM} \\ &= 6\mathbf{a} + \frac{1}{3}(6\mathbf{b} - 6\mathbf{a}) \\ &= 6\mathbf{a} + 2\mathbf{b} - 2\mathbf{a} \\ &= 4\mathbf{a} + 2\mathbf{b}\end{aligned}$$

8aii

$$\begin{aligned}\overrightarrow{LN} &= \overrightarrow{LB} + \overrightarrow{BN} \\ &= \frac{1}{3}\overrightarrow{OB} + (-\overrightarrow{AM}) \\ &= \frac{1}{3}(6\mathbf{b}) - (2\mathbf{b} - 2\mathbf{a}) \\ &= 2\mathbf{a} \\ &= \frac{1}{3}(6\mathbf{a}) \\ &= \frac{1}{3}\overrightarrow{OA} \\ k &= \frac{1}{3}\end{aligned}$$

8b

$$\begin{aligned}\overrightarrow{AB} &= \begin{pmatrix} -5 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} -6 \\ -1 \end{pmatrix}\end{aligned}$$

Since $\overrightarrow{XY}$ is parallel to $\overrightarrow{AB}$,

$$\frac{m}{-2} = \frac{-1}{-6}$$

$$m = -\frac{1}{3}$$

$$(ai) \quad \overrightarrow{BN} = a - \frac{1}{4}b$$

$$(aii) \quad \overrightarrow{OC} = a + b$$

$$(aiii) \quad \overrightarrow{BM} = \frac{1}{2}(a + b) - b$$

$$\frac{1}{2}a - \frac{1}{2}b \quad \text{or} \quad \frac{1}{2}(a - b)$$

$$(bi) \quad \overrightarrow{BQ} = -b + \frac{3}{4}a + \frac{13}{16}b$$

$$= \frac{3}{4}a - \frac{3}{16}b$$

$$= \frac{3}{4}\left(a - \frac{1}{4}b\right)$$

Since $\overrightarrow{BQ} = \frac{3}{4}\overrightarrow{BN}$, BQ//BN.

With B as the common point, B, Q and N are collinear.

$$(bii) \quad 4:3$$